$105/100$

MATHEMATICS 2210

Midterm 3, Fall semester 2012

Name: KEY

1. (15 pts) Find critical points of $f(x, y) = y^3 - 3y + x^3 - 3x$. Which ones are maxima, minima or saddles?

$\nabla f = (3x^2 - 3, 3y^2 - 3) = (0,0) \Rightarrow x^2 = 1$ or $y^2 = 1 \Rightarrow (\pm 1, \pm 1)$ 4 critical pts

$f_{xx} = 6x \quad f_{yy} = 6y \quad f_{xy} = 0 \quad D = 36xy$

$D(1,1) = 36 > 0 \quad f_{xx}(1,1) = 6 > 0$ MIN

$D(1,-1) = -36 < 0$ Saddle

$D(-1,1) = -36 < 0$ Saddle

$D(-1,-1) = 36 > 0 \quad f_{xx}(-1,-1) = -6 < 0$ MAX

2. (15 pts) Find maxima and minima of $f(x, y, z) = x + 2y + 3z$ subject to the constraint $4x^2 + y^2 + 9z^2 = 1$.

$\nabla f = (1, 2, 3) \qquad \nabla g = (8x, 2y, 18z)$

$$\begin{cases} 1 = \lambda 8x \\ 2 = \lambda 2y \\ 3 = \lambda 18z \\ 4x^2 + y^2 + 9z^2 = 1 \end{cases}$$

$8x = 1/\lambda = y = 6z$

$4x^2 + (8x)^2 + 9\left(\frac{4}{3}x\right)^2 = 1$

$x^2(4 + 64 + 16) = 1$

$x = \pm \frac{1}{\sqrt{84}} \Rightarrow \left(\frac{1}{\sqrt{84}}, \frac{8}{\sqrt{84}}, \frac{4}{3\sqrt{84}}\right)$

$\left(\frac{-1}{\sqrt{84}}, \frac{-8}{\sqrt{84}}, \frac{-4}{3\sqrt{84}}\right)$

$f\left(\frac{1}{\sqrt{84}}, \frac{8}{\sqrt{84}}, \frac{4}{3\sqrt{84}}\right) = \frac{1 + 16 + 4}{\sqrt{84}} = \frac{\sqrt{21}}{2}$ MAX

$f\left(\frac{-1}{\sqrt{84}}, \frac{-8}{\sqrt{84}}, \frac{-4}{3\sqrt{84}}\right) = -\frac{\sqrt{21}}{2}$ MIN

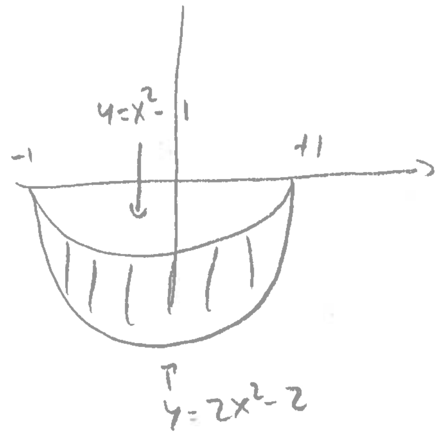
3. (15 pts) Compute $\int\int_S (x^2-1)dA$ where S is the region bounded by $y = x^2 - 1$ and $y = 2x^2 - 2$.

$$\int_{-1}^{1}\int_{2x^2-2}^{x^2-1}(x^2-1)\,dy\,dx = \int_{-1}^{1}(x^2-1)y\Big|_{2x^2-2}^{x^2-1}dx$$

$$=\int_{-1}^{1}(x^2-1)(x^2-1-2x^2+2)\,dx = \int_{-1}^{1}(x^2-1)(-x^2+1)\,dx$$

$$=\int_{-1}^{1}-x^4+2x^2-1\,dx = \left[-\frac{x^5}{5}+\frac{2}{3}x^3-x\right]_{-1}^{1}$$

$$=\left[-\frac{1}{5}+\frac{2}{3}-1\ -\frac{1}{5}+\frac{2}{3}-1\right] = -\frac{16}{15}$$



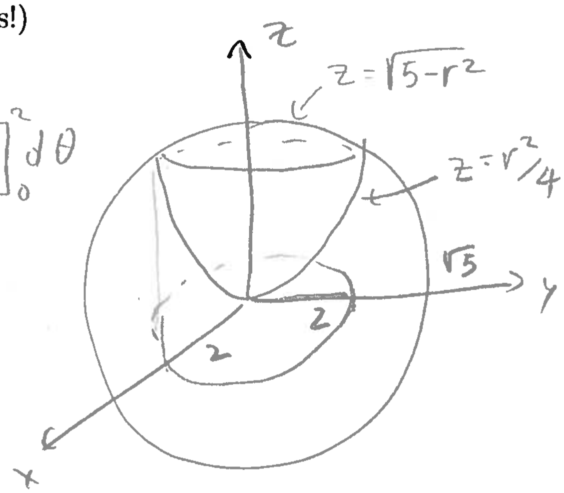
$x^2+y^2+z^2=5$

4. (20 pts) Compute the volume of the solid bounded above by the sphere $r^2+z^2=5$ and below by the paraboloid $r^2=4z$. (Use cylindrical coordinates!)

$x^2+y^2=4z$

$$\int_0^{2\pi}\int_0^{2}\left(\sqrt{5-r^2}-\frac{r^2}{4}\right)r\,dr\,d\theta = \int_0^{2\pi}\left[-\frac{1}{3}(5-r^2)^{3/2}-\frac{r^4}{16}\right]_0^2 d\theta$$

$$=\int_0^{2\pi}\left(-\frac{1}{3}-1+\frac{1}{3}5^{3/2}\right)d\theta = 2\pi\left(\frac{5^{3/2}-1}{3}-1\right)$$



$4z = x^2+y^2 = 5-z^2$

$z^2+4z-5=0$

$(z-1)(z+5)=0$

$\underline{z=1}, \cancel{5}\ r^2=4$

5. (15 pts) Compute the work done by the force field $F(x,y) = x^2\,\mathbf{i} + xy\,\mathbf{j}$ in moving a particle along the curve $x^2 = y^3$ between $(1,1)$ and $(8,4)$.

$r(t) = (t^3, t^2) \quad 1 \le t \le 2 \qquad dr = (3t^2, 2t)\,dt$

$F(r(t)) = (t^6, t^5) \qquad W = \int F\cdot dr = \int_1^2 (t^6, t^5)\,(3t^2, 2t)\,dt =$

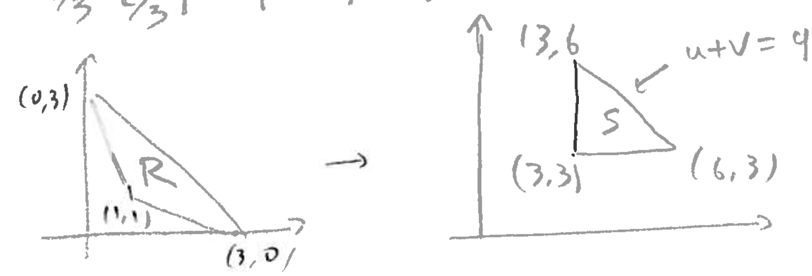
$= \int_1^2 3t^8 + 2t^6\,dt = \left[\frac{3}{9}t^9 + \frac{2}{7}t^7\right]_1^2 = \frac{2^9}{3} + \frac{2^8}{7} - \frac{1}{3} - \frac{2}{7}$

$(0,3)\ (3,0)$

6. (20 pts) Let R be the triangle with vertices $(0,0)$, $(1,0)$ and $(1,1)$. Using the change of variables formula for $u = 2x + y$ and $v = x + 2y$, express the integral $\int\int_R \left(\frac{2x+y}{x+2y}\right) dA$ as an integral in du and dv. **(Setting up the correct integral will earn full credit. Computing the resulting integral is worth 5 bonus points.).**

$x = \frac{2u - v}{3} \qquad J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{vmatrix} = \frac{4}{9} - \frac{1}{9} = \frac{3}{9} = \frac{1}{3}$

$y = \frac{2v - u}{3}$



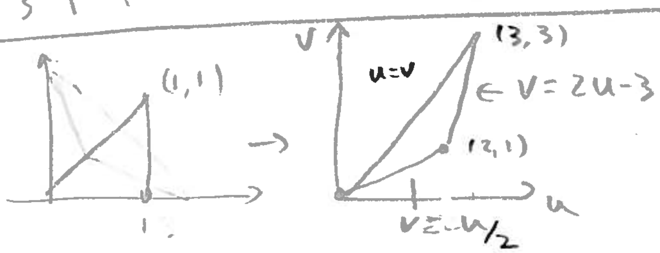
$\int\int_R \left(\frac{2x+y}{x+2y}\right) dA$

$= \int_3^6 \int_3^{9-v} \frac{u}{v}\cdot\frac{1}{3}\,du\,dv \Leftarrow$ Full credit

$= \frac{1}{3}\int_3^6 \frac{u^2}{2v}\Big|_3^{9-v} dv = \frac{1}{3}\int_3^6 \frac{(9-v)^2}{2v} - \frac{9}{2v}\,dv = \frac{1}{3}\int_3^6 \frac{v^2 - 18v + 72}{2v}\,dv = \frac{1}{3}\left[\frac{v^2}{4} - 9v + 36\ln v\right]_3^6$

$= \frac{1}{3}\left(\frac{36}{4} - 54 + 36\ln 6 - \frac{9}{4} + 27 - 36\ln 3\right) = -\frac{27}{4} + 12\ln 2 \Leftarrow +5$ pts

OR



$\int_0^1 \int_v^{2v} \frac{u}{v}\cdot\frac{1}{3}\,du\,dv + \int_1^3 \int_v^{\frac{v+3}{2}} \frac{u}{v}\cdot\frac{1}{3}\,du\,dv$

Full credit